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Research paper

Multi-view Deep Embedded Clustering: Exploring a new dimension of air pollution

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ABSTRACT

Clustering is essential for uncovering hidden patterns and relationships in complex datasets. Its importance reveals when labeled data is scarce, expensive, time-consuming to obtain. Real-world applications often exhibit heterogeneity due to the diverse nature of the encapsulated data. This heterogeneity poses a significant challenge in data analysis, modeling, and makes traditional clustering methods ineffective. By adopting a hybrid architecture based on two promising techniques, multi-view and deep clustering, our method achieved better results, outperforming several existing methods including K-means, deep embedded clustering, deep clustering network, deep embedded K-means among many others. Multiple experiments conducted across diverse publicly accessible datasets validate the effectiveness of our proposed method based on well established evaluation metrics such as Accuracy and Normalized Mutual Information (NMI). Furthermore, we applied our method on the air pollution data of Luxembourg, a country with sparse sensor coverage. Our method demonstrated promising results, and unveil a new dimension that pave way for future work in air pollution's level prediction and hotspots detection, crucial steps towards effective pollution reduction strategies.

1. Introduction

In machine learning, clustering (Dornaika and El Hajjar, 2023, 2024; El Hajjar et al., 2024) is used to group similar data points, enabling the discovery of inherent structures in datasets. Several clustering methods are commonly employed, with the K-means clustering being one of the oldest and most well-known of these methods. Its popularity comes from its simplicity and ease of implementation, making it a favored choice for many. However, K-means clustering heavily relies on initialization, which can sometimes hinder its effectiveness. Moreover, it tends to struggle when attempting to cluster data of certain shapes, adding another layer of complexity to its use. Another prominent approach in the field is the spectral clustering method. This method is often used to enhance the results obtained from K-means clustering. It introduces a spectral projection matrix that maps data points into a separable space prior the final clustering step, offering a new dimension of analysis. This method is based on the principle of the Laplacian graph. However, similar to K-means, it also has a degree of dependence on initialization, which can pose challenges in certain scenarios. In practical applications, the limitations of K-means and spectral clustering become evident when dealing with complex datasets such as those encountered in air pollution studies. K-means, for example, may misclassify clusters when the data does not conform to simple geometric shapes, leading to inaccurate identification of pollution hotspots. Spectral clustering, while useful in enhancing cluster separation, struggles with initialization issues and may produce suboptimal results when applied to high-dimensional or multi-density data, such as that found in urban air pollution monitoring. To address these issues, multi-view clustering methods had been proposed. Multi-view clustering methods (Chao et al., 2021) refer to algorithms and techniques used in machine learning and data analysis to group data points into clusters based on multiple sets of features or "views" of the data. Each view represents a different perspective or aspect of the data, and by considering multiple views simultaneously, these methods aim to improve clustering accuracy and robustness. Nowadays, these methods are becoming more prominent in various fields. While clustering consists of dividing the dataset into different groups with high intra-class similarity, the use of distinct views for the same dataset lead to improved clustering results. In fact, using multiple representation or

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views for the same dataset give complementary information about the distinct groups. It could also reduce the noise present in some views, making the use of multiple views more efficient for better clustering results. In real-world datasets, data may exhibit heterogeneity across different views. Multi-view clustering can effectively handle such heterogeneity by accommodating diverse data types or representations within a single model. Moreover, multi-view clustering naturally incorporates supervision or side information from one or more views, enabling semi-supervised or weakly supervised clustering, and enhancing clustering performance in scenarios where labeled data is limited. This empowers us with an interesting technique to address one of the largest environmental threat for human health — air pollution.

Cities all over the world are changing significantly as a result of various factors, such as quick urbanization, new means of communication, advancements in industry and technologies, and continuous reliance on fuel-based transportation. The benefits of technology and urbanization are undeniable, yet they come at a steep cost, with air pollution being a major consequence of this progress. Air pollution is recognized by the World Health Organization (WHO) as the world's largest environmental stressor for human health. Globally, around 7 million people die annually due to air pollution (WHO Authors, 2023), and more than 200 million years of life lost (YLL). Air pollution is also perceived among Europeans as the second biggest environmental concern with a share of death percentage between 3.5% to 6% in the overall all-cause death rate in Europe (Juginović et al., 2021). Moreover, according to Eurostat, the percentage of population reporting they suffered from asthma reached a percentage as high as 12% and a percentage as high as 9% for other chronic respiratory diseases in 2019 (Eurostat Authors, 2023). One of the main air pollutants that is confirmed to have a direct relation with the diseases caused by the ambient air pollution is the Nitrogen dioxide (NO_2). NO_2 is a reddish-brown gas which results from the reaction of Nitrogen Oxide (NO) with some volatile organic compounds. While the NO emission sources are well known to be mainly the vehicle traffic, industrial sector, power plant activity, heating and airports, countries all over the world are always facing a problem keeping the NO_2 gas level below the warning threshold set by the WHO. Moreover, the NO_2 gas has a potential to produce secondary pollutants such as Peroxyacyl Nitrates (PANs), nitric acid, and ozone (O3), which contribute to the formation of smog, acid rain, and the greenhouse effect, respectively (Schmitz et al., 2019). Therefore, people are prone to suffer from poor health effects from pollutants if they live or work close to air pollution sources such as congested roadways or industries that create air pollutants (U.S. Environmental Protection Agency). The spatial analysis of air quality is of paramount importance for assessing health impacts and effectively managing air pollution, especially in regions characterized by elevated ambient pollution concentrations, commonly referred to as "air pollution hotspots" (Gately et al., 2017). However, air pollution levels are subject to considerable variability, which can complicate hotspot identification. Pollutants may originate from sources in regions different from the area where high pollution levels are observed, as pollutants can be transmitted across space, for instance the wind can disperse the NO_2 pollutant in 30 min (Richmond-Bryant et al., 2018). Therefore, it becomes imperative to adopt a comprehensive approach that accounts for different factors and features to detect air pollution hotspots accurately.

Clustering emerges as a prominent technique for identifying air pollution patterns, yet it can be categorized into two distinct groups. The first group comprises algorithms utilizing global parameters and a single minimum threshold value to differentiate between dense and non-dense areas. However, when clusters in different regions of the feature space possess significantly different densities or when clusters with varying density levels are nested, determining an appropriate threshold becomes challenging. A single-density threshold may lead to improper partitioning, generating small non-significant clusters that do not represent dense regions when the threshold is set too high, or discovering a few large regions that are no longer dense when the threshold is set too low. Consequently, applying such algorithms to multi-density datasets, such as air pollution data, may not yield satisfactory results. The second group consists of algorithms employing multiple minimum threshold values. These algorithms endeavor to detect multiple pattern distributions of diverse densities, aiming to differentiate between various density regions, which may or may not be nested and often exhibit non-convex shapes. Unlike single-density threshold algorithms, they autonomously estimate the number of threshold values to optimally identify different density regions without any prior knowledge of the data. Although these algorithms generally achieve good partitioning, they suffer from the drawback of high computational cost.

To overcome these challenges, deep clustering emerges as a viable approach to studying the complex features of the air pollution data. By leveraging the power of deep learning techniques, deep clustering can learn intricate representations from the data, capturing underlying patterns and structures that might be difficult to understand with traditional clustering algorithms. The capability of deep clustering to automatically extract relevant features from air pollution datasets, offers a promising solution to enhance air pollution patterns detection and achieve more accurate data partitioning. Additionally, deep clustering methods (Hwang and Kim, 2020; Jenkins et al., 2021; Jiang et al., 2019) often exhibit better generalization abilities, making them well-suited for handling diverse and complex datasets.

In this paper, we propose a novel deep multi-view clustering method for detecting interesting patterns in heterogeneous data, capitalizing on the advantages of deep clustering to address the challenges posed by multi-density, and enhancing the overall precision. Our method had been first tested to some public datasets including REUTERS-10K, 20NEWS, and RCV1-10K in order to demonstrate its efficiency. Having demonstrated superiority to other state of the arts methods, we turned to apply our method to air pollution, a real-world problem. Moreover, we incorporated additional features beyond air pollution data in order to acknowledge the correlation between pollutants and the diverse land-use features, and to allow for a more robust characterization of air pollution patterns, considering the potential spatial transmission of pollutants. Through experimentation, we demonstrate the robustness and adaptability of our method, providing valuable insights for environmental monitoring, and urban planning. The results of this research contribute significantly to the field of environmental science and pave the way for more informed and effective pollution mitigation strategies to combat the pressing global threat of air pollution and its associated health impacts.

The primary contributions of this paper can be outlined as follows.

1. We introduce a new deep multi-view clustering method for detecting different patterns within heterogeneous datasets, building upon the existing Deep Embedded K-Means Clustering method (Guo et al., 2021).
2. By integrating additional land-use and neighbors factors, our solution offers a comprehensive and accurate method to identify air pollution patterns and behaviors, and acknowledges potential cross-regional influences.
3. We create two distinct views of the data by incorporating the features generated from two different deep learning models. This multi-view architecture enriches the representation of the data, thus strengthening the clustering results.
4. We conduct accurate tests and evaluations on public datasets to demonstrate the efficiency of the proposed multi-view deep technique. Additionally, we tested our method in the context of the Grand Duchy of Luxembourg to showcase its accuracy in identifying the different air pollution patterns, further reinforcing its potential for real-world applications.

The remainder of this paper is structured as follows: In Section 2, we provide a review of related work in air pollution clustering and prediction, exploring existing methods, their strengths and limitations.

Section 3 details the methodology of our proposed multi-view deep clustering method for air pollution patterns identification, incorporating diverse datasets to enhance accuracy. In Section 4, we conduct experiments on public datasets, demonstrating the efficiency of our method. Section 5 discusses the details of the application and evaluation of our method on air pollution dataset. Finally, Section 6 concludes the paper.

2. Related work

Applying clustering techniques to real-world applications such as the air pollution presents several challenges due to the inherent complexity and variability of data. Real-world data often exists in high-dimensional spaces, which can lead to the curse of dimensionality. Clustering algorithms may struggle to effectively partition data in such spaces, requiring dimensionality reduction techniques or specialized clustering algorithms designed to handle high-dimensional data. Although dimensionality reduction play crucial role in simplify data, they imply loss of information, parameter complexity, and computational cost. They often yield non-intuitive interpretations, are sensitive to outliers and noise, and have high computational complexity. Our method addresses this by employing a deep learning approach using autoencoders, which can learn non-linear representations of high-dimensional data without losing critical information. This allows us to preserve the structure of the data in a lower-dimensional embedding space while being robust to noise and outliers.

Moreover, another challenge when applying clustering techniques to a real-world application is the multi-density of the dataset. In clustering, multi-density refers to the complexity encountered when attempting to cluster datasets that exhibit varying densities across different regions and dimensions. Unlike traditional clustering problems where data points are uniformly distributed, multi-density datasets pose a unique challenge due to the presence of dense clusters alongside sparse ones. This creates difficulties in defining appropriate cluster boundaries and accurately identifying clusters of different densities. Traditional clustering algorithms such as K-means struggle to handle such scenarios effectively, often leading to poor results. Addressing the multi-density challenge requires the development of advanced clustering techniques capable of adapting to the varying density levels within the data, such as density-based clustering algorithms like DBSCAN or hierarchical clustering methods. These approaches can better capture the inherent structure of multi-density data sets by allowing for flexible cluster shapes and accommodating regions of different densities. However, due to their limitations in handling high-dimensional data with complex spatial and temporal features, DBSCAN and hierarchical clustering may struggle with real-world problems and become ineffective in capturing the intricate relationships and variations. Our method overcomes this by integrating multi-view clustering with an orthonormal transformation matrix derived from K-means. This allows us to handle the varying densities of clusters in high-dimensional spaces more effectively, ensuring that both dense and sparse clusters are identified without sacrificing performance in complex, real-world datasets.

The Density-based Sequential Pattern Mining (DSPM) approach, described in Cesario et al. (2017a), intended to explain mobility patterns from GPS data. The procedure consists of two fundamental steps: (1) identifying urban dense regions of interest, which are areas that are more frequently traveled through; and (2) capturing mobility patterns between these regions. The study applies this approach to a real-life GPS dataset, tracking the movements of taxis in the urban area of Beijing, showcasing its practical application and potential for understanding urban mobility. The authors also present a comprehensive validation methodology to assess the accuracy and quality of the detected dense regions and trajectory patterns. Although the approach relies on the DBSCAN algorithm for detecting dense regions, the authors acknowledge the possibility of enhancing the method by considering multi-density clustering analysis, which could also identify lower-density but homogeneous regions, providing a more comprehensive view of urban mobility patterns. Our multi-view approach offers a more comprehensive solution by capturing both high-density and low-density regions. By fusing representations from two different views of the data, we ensure that even homogeneous, lower-density regions are identified accurately, improving overall cluster detection across diverse regions.

In Deng et al. (2022), the authors propose a novel method for short-term ozone forecasting using clustering-based spatial transfer learning. The approach utilizes k-means clustering to identify similar observation stations, which are then jointly trained to establish a base model for spatial transfer learning. This transfer learning enables more accurate ozone concentration predictions for target observation stations, even with limited data available for those specific locations. The study evaluates the approach using historical data from observation stations in Germany and demonstrates its superiority in ozone exceedance forecasting compared to traditional chemical transport models and popular machine learning approaches. However, considering that the sensor stations cover a large area and might exhibit multi-density characteristics, the authors suggest that employing multi-density clustering algorithms could further enhance the model's performance, potentially improving the accuracy of ozone predictions in regions with varying density patterns. Our method's ability to handle multi-density datasets through advanced clustering techniques based on deep learning ensures more accurate predictions. By adapting our clustering to varying air pollution levels, we improved the forecasting accuracy in regions with different pollution levels, addressing the shortcomings of traditional K-means-based approaches.

In the paper Khan et al. (2021), the authors propose an approach to predict high-resolution electric consumption trends at finely resolved spatial and temporal scales. The method consists of two main steps. Firstly, historical electric consumption data at the apartment-level are collected and clustered. Subsequently, the clusters are aggregated based on the consumption profiles of consumers. The clustering analysis employs the k-means algorithm, while forecasting models are derived using two deep learning techniques: long short-term memory unit (LSTM) and gated recurrent unit (GRU). The experimental evaluation was conducted using electricity consumption data obtained from residential buildings in an urban area of South Korea. Notably, a comparative analysis was performed, comparing the approach's performance with state-of-the-art machine learning models and various deep learning variants. The results demonstrated good prediction accuracy at both the building- and floor-level, showcasing the efficacy of the proposed method in electric consumption forecasting. However, one limitation of the approach is that the clustering of consumption profiles does not consider features related to the location of apartments, buildings, and floors. This aspect could be further enhanced by incorporating multi-density clustering techniques. By utilizing multi-density clustering, buildings in the same city area, constructed in similar years, and with similar characteristics, could be grouped together. This enhancement may lead to more informative cluster formations, providing additional insights into electric consumption trends and facilitating improved decision-making for energy management and urban planning. By incorporating spatial features (i.e., land use, etc.) into our multi-view clustering approach, our method enables more accurate predictions based on shared characteristics. This leads to better insights into air pollution trends and more effective decisions.

In Kolevatova et al. (2021), the authors present a comprehensive workflow comprising five essential steps for analyzing the impacts of changes in land cover, such as deforestation or urbanization, on the local climate. The workflow encompasses data preprocessing, feature extraction, machine learning training, performance evaluation, and explainable artificial intelligence. The primary objective is to investigate the relationship between land cover changes and temperature variations. The study utilizes machine learning models trained to understand this correlation between changes in land cover and temperature. Subsequently, explainable artificial intelligence is employed to interpret and analyze the effects of different land cover alterations on temperature. This interpretability aspect provides valuable insights into the underlying mechanisms driving the climate change response to land cover modifications. The experimental results demonstrate that random forest outperforms other machine learning methods, including linear regression, as proposed in the literature for discovering the relationship between land cover changes and temperature variations. The workflow offers a robust framework for assessing the impact of land cover transformations on local climate conditions, providing a valuable tool for environmental researchers, policymakers, and urban planners to make informed decisions regarding land use and its consequences on regional climate patterns. Despite the good performance of the model, it relies on models like random forests that may not fully capture complex patterns in multi-density or multi-dimensional data. Our deep learning-based approach allows for the integration of multi-dimensional features and land-use patterns, providing more accurate interpretations of how land cover changes impact local climates. This offers better insights compared to traditional methods that may overlook these complex relationships.

In Cesario et al. (2017b), a methodology is introduced to discover behavior rules, correlations, and mobility patterns of visitors at large-scale public events using social media posts analysis. The multi-step approach involves the detection of air pollution at regions of interest, the collection of geo-tagged items related to the events, gathering user trajectories from posts, and the discovery of touristic mobility patterns. The methodology is tested through two case studies: an analysis of mobility patterns of Instagram users who visited EXPO 2015 and behavior modeling of geo-tagged tweets from users attending the 2014 FIFA World Cup. The results demonstrate reasonable predictive accuracy, showcasing the potential of social media data in understanding visitor behavior and mobility patterns during major public events. The mentioned social media analysis method relies on large-scale data but may miss out on capturing more intricate patterns of mobility or air pollution due to the limitations of single-view clustering techniques Our multi-view clustering method, by fusing information from multiple perspectives (e.g., land-use, mobility data, pollution levels, etc.), enhances the detection of finer patterns. This provides a more detailed and accurate understanding of visitor behavior and mobility, especially in pollution-sensitive regions.

A recent study in Cai et al. (2022) introduces a state-of-the-art deep learning recommendation model, DeepCGSR, which enhances recommendation accuracy by integrating fine and coarse-grained sentiment analysis from user reviews with user-item rating matrix factorization. This approach addresses issues like data sparsity and the cold-start problem, offering significant improvements over existing models. In Zhu et al. (2023a), the authors present a blockchain-based service architecture designed to enhance the traceability and protection of original achievements, such as patents and copyrights. This innovative approach leverages the core features of blockchain technology to secure and validate intellectual property transactions, offering a robust solution that outperforms existing methods in terms of effectiveness and reliability. Moreover, the authors in Zhu et al. (2024) introduce a novel discrete conformable fractional grey system model designed for forecasting carbon dioxide emissions. This model enhances traditional grey-based forecasting methods by incorporating conformable fractional calculus, resulting in superior predictive accuracy. Tested on CO2 emissions data from Germany, Japan, and Thailand, the model outperformed several benchmark models, offering valuable insights into emission trends and aiding in the development of more effective environmental policies. In Zhu et al. (2023b), the authors investigate the impact of informational cascades on consumer behavior in online flash shopping environments. Their study reveals that consumers' tendency to imitate others and their perceptions of value significantly influence purchase intentions during flash sales. By examining the psychological processes behind these decisions, the research provides insights that can help e-commerce retailers craft strategies to boost impulse purchases, emphasizing the importance of consistent product reviews, pre-sale indicators, and strategic information management. While these approaches excel in specific areas, our method generalizes to complex datasets, integrating multiple views of data and handling varying densities within the data space. By leveraging a deep learning-based framework, we provide a more robust solution capable of capturing intricate relationships in heterogeneous datasets, offering superior performance in real-world environmental applications.

Moreover, in Canino et al. (2022), a predictive approach based on spatial analysis and regressive models is presented, aiming to uncover spatio-temporal predictive epidemic patterns from infection and mobility data. The algorithm consists of several steps, starting from the detection of epidemic air pollution levels (urban areas with dense infection events) and mobility air pollution levels (urban regions with dense mobility traces). It proceeds to discover epidemic patterns among these air pollution levels, and finally, processes each epidemic air pollution levels to extract air pollution-specific epidemic forecasting models. The validation of the approach is conducted using real-world data from mobility and COVID-19 infections in Chicago. The paper focuses specifically on high-density air pollution in the analysis and employs the DBSCAN algorithm to detect epidemic air pollution level. However, the authors suggest that incorporating other multi-density-based clustering algorithms could further enhance the approach's performance, providing a more comprehensive understanding of epidemic patterns across different density regions. The study showcases the potential of spatial analysis and predictive modeling in understanding the spatio-temporal dynamics of epidemics, contributing to informed decision-making and effective public health management during disease outbreaks. Despite the high efficiency of the existing methods for air pollution levels and patterns detection, they suffer from limitations such as inadequate handling of multi-density regions, lack of interpretability, and suboptimal predictive accuracy, often neglecting the integration of crucial data features, such as land use and other relevant types. These shortcomings have motivated us to develop a novel deep learning-based approach that comprehensively considers diverse data features. By incorporating features like land use and pollution data, our model achieves more accurate and actionable insights for detecting air pollution patterns.

Existing methods for detecting air pollution levels and patterns face several limitations:

- Handling of Multi-Density Regions: Many current approaches struggle with datasets characterized by varying densities, often leading to poor clustering performance in regions with complex density distributions.
- Interpretability Challenges: Current techniques often lack transparency, making it difficult to understand the relationship between data features and clustering results.
- Predictive Accuracy: Traditional methods may not fully leverage the complex, multi-dimensional nature of real-world data, resulting in suboptimal accuracy for predicting pollution patterns.

To address these challenges, our proposed method introduces a novel approach leveraging deep learning to enhance clustering performance. By integrating multi-view representations and employing advanced optimization techniques, our method aims to improve the handling of multi-density data, enhance interpretability, and provide more accurate predictive insights.

3. Proposed solution

To overcome the challenges of previously mentioned methods, our paper introduces a novel deep multi-view clustering method that by leveraging the power of deep learning, it captures complex features in the data, contributing to enhanced precision multi-density clustering applications and offering cluster-structure information. Our method is inspired from the idea of the Deep Embedded K-Means Clustering model presented in Guo et al. (2021).

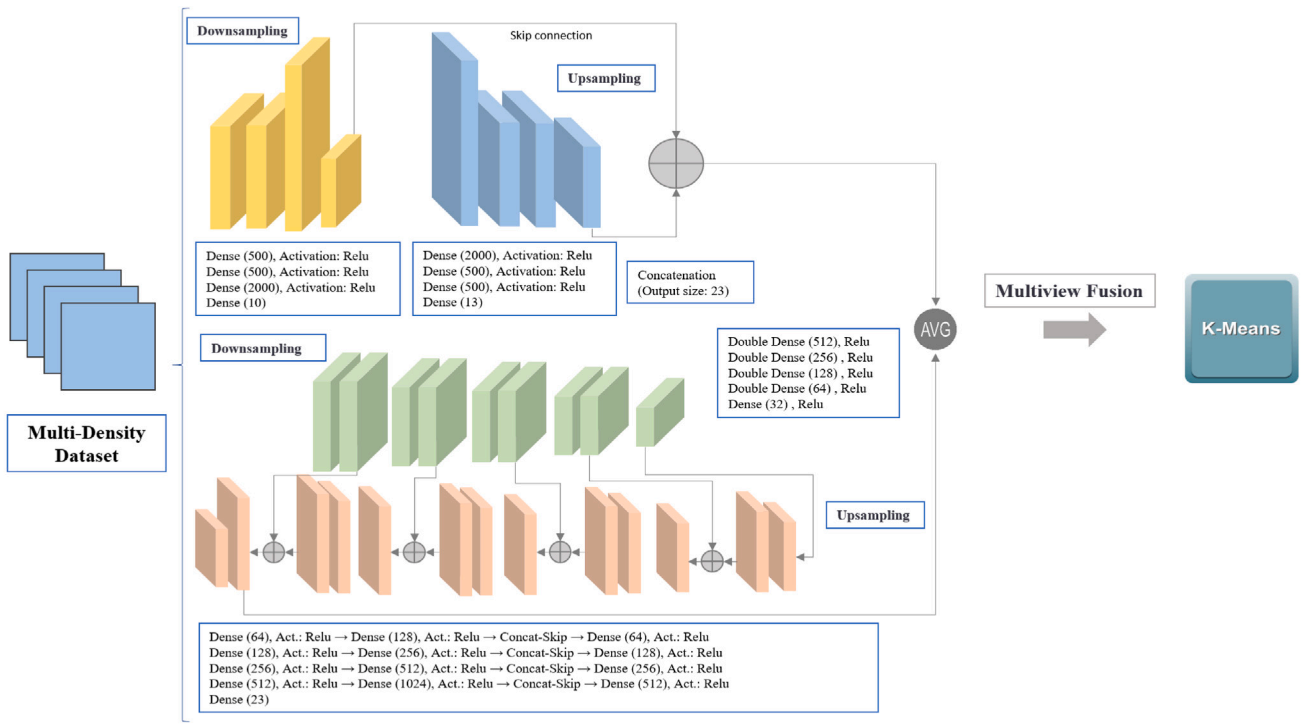

Fig. 1. Description of the proposed method.

3.1. Review of the Deep embedded K-means clustering method (DEKM)

The idea behind the paper called "Deep embedded K-means clustering", is to extend the traditional K-means algorithm by introducing a deep version called Deep Embedded K-means (DEKM). The goal is to enhance the clustering performance by utilizing an autoencoder to generate an embedding space. The autoencoder is used to transform the input data into a lower-dimensional representation, which serves as the embedding space. However, the initial embedding space generated by the autoencoder may not be sufficiently discriminative for effective clustering. To address this limitation, the authors propose further transforming the embedding space using an orthonormal matrix comprising eigenvectors derived from the within-class scatter matrix of K-means. These eigenvectors are arranged in ascending order based on their eigenvalues, with each eigenvalue indicating the extent to which its corresponding eigenvector contributes to the cluster-structure information in the new space. Notably, the last eigenvector contributes the least. To amplify the cluster-structure information along the direction of the last eigenvector, the authors adopt a greedy optimization method to refine the representation further by discarding the decoder part of the autoencoder. The optimization process aims to minimize the entropy, which is consistent with the loss function used in K-means. This iterative optimization alternately updates the representation learning and clustering steps until a convergence criterion is met. In summary, the main idea of the aforementioned paper is to develop a deep learning-based method (DEKM) that combines autoencoder-based representation learning with K-means clustering, aiming to improve the clustering performance and reveal the underlying cluster-structure information more effectively.

3.2. Methodology

In this paper, we proposed a novel deep multi-view clustering method for multi-density clustering applications. The main objective of our solution is to address the limitations of traditional clustering algorithms when dealing with multi-density datasets and to enhance the accuracy, efficiency, and performance of the model. Our method builds upon the concept of Deep Embedded K-Means Clustering (DEKM), leveraging the power of deep learning to learn complex features and representations from different views of the data. The key innovation of our approach lies in the utilization of two different autoencoders to create distinct views of the data. Each autoencoder is tailored to a particular aspect of the data, thus responsible for extracting meaningful features from a specific set of data attributes, enhancing the representation learning process. The first model adopted in our method is the same as used in the Deep Embedded K-Means Clustering method. Furthermore, as shown in Fig. 1, we integrate another model, adhering to the concept of the Unet model (Ronneberger et al., 2015), which mainly consists of an encoder and a decoder along with skip connections units. The use of the skip connections layers enable fine-grained feature propagation, significantly enhancing performance by making it possible to the network to capture both low-level and high-level features effectively. This model is used to create the second view of our dataset. Once we have obtained the encoded representations from both autoencoders, we take the average of their outputs to create a fused representation. By combining the two views in this manner, we aim to integrate complementary information and produce a more comprehensive and robust representation of the data. Next, we introduce our deep multiview clustering algorithm, which is used the fused features of the two different views. By applying an orthonormal transformation matrix containing eigenvectors from the within-class scatter matrix of K-means to the multiview embedding space, we reveal the cluster-structure information in a new space. This transformation allows us to overcome the multi-density challenges and obtain more accurate clustering results. Furthermore, we adopt an iterative optimization approach that alternately updates the representation learning and clustering steps. This joint optimization process ensures that the representation aligns with the clustering objectives, ultimately leading to improved efficiency. The description of the proposed model is illustrated in Fig. 1.

Fig. 2 illustrates the proposed model architecture. To show the detailed layer shapes, we are illustrating an example of the model applied to the air pollution dataset.

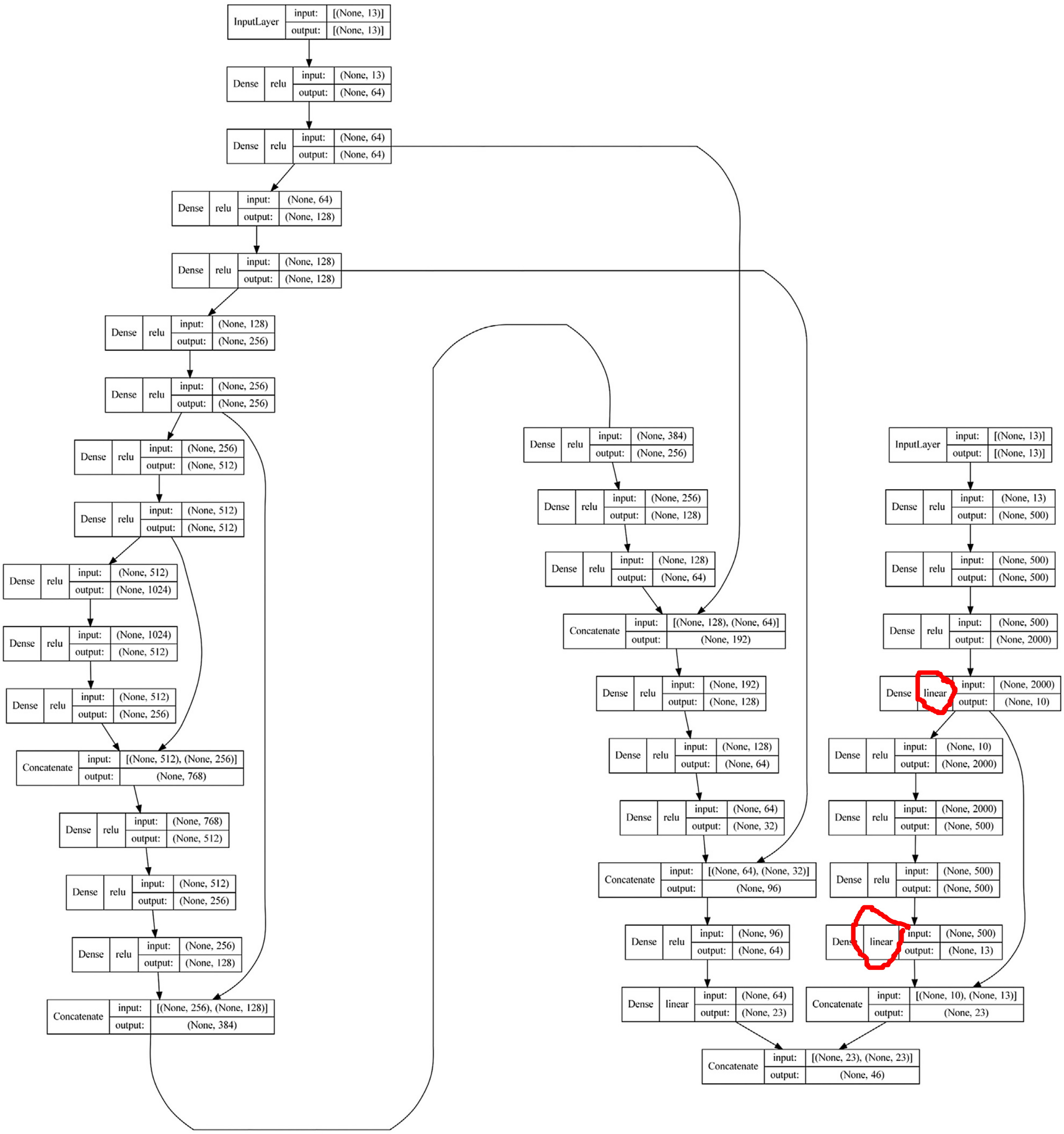

Fig. 2. Proposed model architecture.

Furthermore, our proposed solution integrates a multiview learning approach with different network structures tailored to each view of the data. We utilized two different autoencoders, where:

- The first autoencoder follows a deep fully connected structure, similar to the architecture used in Deep Embedded K-Means (DEKM). The encoder has layers with dimensions that progressively reduce, such as 500-500-2000. This structure was chosen for its ability to capture and compress the information in the input space effectively, focusing on key latent features. The decoder mirrors this structure with ReLU activation functions, and Adam optimizer was employed for fast and stable convergence.
- For the second view, we employ a U-Net inspired architecture, which consists of an encoder and decoder with skip connections. This model captures fine-grained and high-level features, enabling the extraction of complementary information to the first autoencoder.

After encoding both views, we fuse the learned representations by averaging their outputs, as described in Eq. (6). This fused embedding is then clustered using K-means, with an orthonormal transformation matrix to improve cluster separation.

Several machine learning (particularly neural networks) concepts and elements have been used to develop the proposed model. In the following sub-sections we give an overview of each of these concepts and elements.

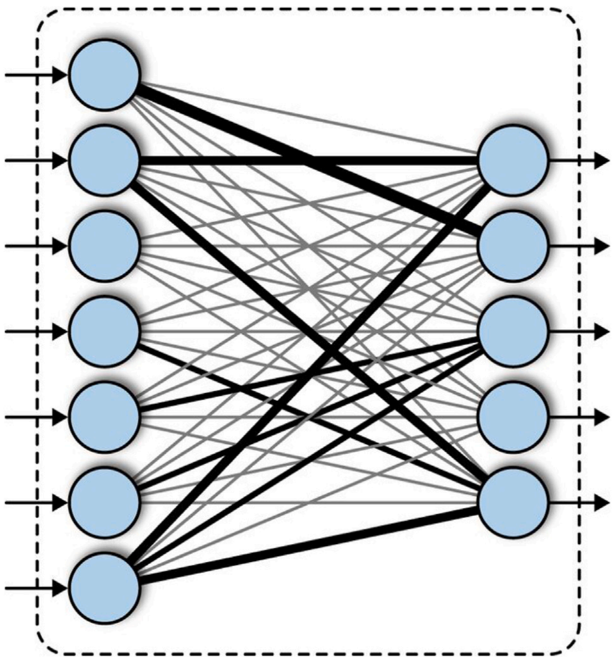

Fig. 3. Visualization of a dense layer.

3.2.1. Dense layer

Dense layer (also known as fully connected layer) is a fundamental component of deep learning architectures. They play a critical role in the network's ability to learn complex patterns from data (LeCun et al., 2015). Structure-wise, it consists of neurons, connections, weight and biases. In this type of layers, every neuron is connected to every neuron in the previous layer. This means that each neuron receives input from all the neurons in the preceding layer and passes its output to all neurons in the next layer. Each of these connections between neurons has an associated weight, and each neuron has a bias term. These parameters are learned during training and are critical for the model's predictive ability. The operation in a dense layer is a linear transformation followed by a non-linear activation function. Mathematically, for a layer with input vector x, the output y is computed as:

$$y = A(W * x + b). \tag{1}$$

where:

$W =$ Weight matrix
$x =$ Input vector
$b =$ Bias vector
$A =$ Activation function

Dense layers help in learning high-level features from the input data. They take the complex patterns detected by previous layers and combine them to make predictions or classifications. Moreover, due to their dense connectivity, dense layers are highly flexible and capable of approximating complex functions. This makes them a key component in models that need to capture intricate patterns in the data, this make it a perfect fit for our case. Fig. 3 shows a visualization of a dense layer (Hollemans, 2024).

The main hyper-parameters that highly affect the performance of a dense layer are the number of neurons, the activate function.

3.2.2. ReLU activation function

Activation functions are a crucial component in neural networks, playing a significant role in determining the model's ability to learn and make complex predictions. They introduce non-linearity into the network, without of wich, a neural network with multiple layers would behave like a single-layer linear model, limiting its capacity to model complex patterns. ReLU (Rectified Linear Unit) is one of the most widely used activation functions in modern neural networks, particularly in deep learning architectures. Its simplicity (thus computationally efficient) and effectiveness have made it a default choice for many applications. It outputs the input directly if it is positive; otherwise, it outputs zero.

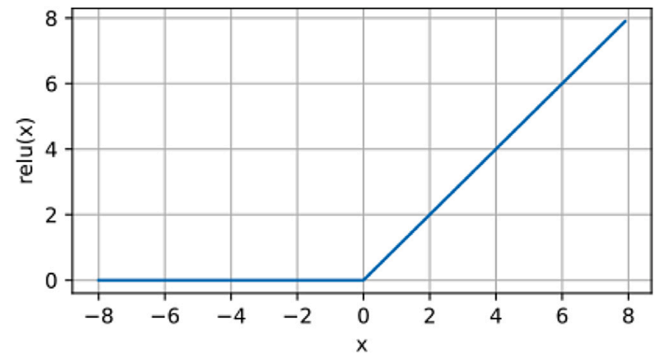

Fig. 4. ReLU activation function.

$$ReLU(x) = \max(0, x). \tag{2}$$

where:

$x =$ Input vector

Fig. 4 shows the ReLU activation function.

Although ReLU appears to be a piecewise linear function, it introduces non-linearity to the network, which is crucial for learning complex patterns. Moreover, ReLU can result in sparse activations, meaning that in any given layer, some neurons may not be activated (output is zero). This sparsity can lead to more efficient models.

3.2.3. Adam optimizer

Adam(Adaptive Moment Estimation) (Kingma and Ba, 2014) is one of the most widely used optimization algorithms in deep learning due to its efficiency, robustness, and adaptability. It combines the advantages of two other popular optimizers: AdaGrad and RMSProp, making it suitable for a wide range of tasks and models. Adam is an extension of stochastic gradient descent (SGD), an optimization method that updates model parameters iteratively to minimize a loss function. It incorporates the concept of momentum, which accelerates SGD in the relevant direction and dampens oscillations. Momentum helps Adam move faster along the gradient and reduces the risk of getting stuck in local minima. It also adapts the learning rate dynamically for each parameter based on the first and second moments of the gradients, leading to more stable and efficient training. We chose to use Adam as the optimzer for our proposed model for its power in combining the strengths of momentum and adaptive learning rates, and its fast convergence and computational efficiency.

3.3. joint optimization framework

3.3.1. representation learning (autoencoders)

In our deep multi-view clustering method, we use two separate autoencoders, each capturing different information from the data. The first autoencoder learns from set of features, while the second captures complementary information from the second set. Let $f_1(\cdot)$ and $f_2(\cdot)$ represent the encoders of the two autoencoders. For each data point $\boldsymbol{x}_i$, we generate two distinct embeddings:

$$h_i^{(1)} = f_1(x_i), \quad h_i^{(2)} = f_2(x_i). \tag{3}$$

To integrate both views into a unified representation, we compute the fused embedding as the average of the outputs from the two autoencoders:

$$h_i = \frac{1}{2}\left(h_i^{(1)} + h_i^{(2)}\right). \tag{4}$$

This step ensures that the complementary information from both views is combined, producing a more comprehensive representation of the data. The autoencoders are trained by minimizing the reconstruction loss for both views. The reconstruction loss for each view measures the difference between the original data and the data reconstructed by the autoencoder:

$$L_1 = \sum_{i=1}^{n} \left(\|x_i - g_1(f_1(x_i))\|^2 + \|x_i - g_2(f_2(x_i))\|^2\right), \tag{5}$$

where $g_1(\cdot)$ and $g_2(\cdot)$ are the decoders corresponding to the encoders $f_1(\cdot)$ and $f_2(\cdot)$. This encourages the autoencoders to learn meaningful, low-dimensional representations from each view while ensuring the reconstruction fidelity of the input data.

3.3.2. clustering in the fused embedding space (K-means objective)

Once we have the fused embeddings h_i from both autoencoders, we apply the K-means clustering algorithm to these embeddings. The objective function of K-means minimizes the distance between the fused embeddings and their corresponding cluster centroids:

$$L_2 = \sum_{j=1}^{k} \sum_{h_i \in C_j} \|h_i - \mu_j\|^2, \tag{6}$$

Here, C_j denotes the set of points assigned to cluster j, and μ_j is the centroid of that cluster. This loss encourages the fused embeddings to form tight clusters around the centroids, improving the clustering performance.

3.3.3. orthonormal transformation using eigenvectors

After clustering the fused embeddings, we apply an orthonormal transformation to further enhance the cluster structure. This transformation uses an orthonormal matrix V, which consists of eigenvectors from the within-class scatter matrix S_w of the K-means clustering. The transformed embedding space y_i is computed as follows:

$$y_i = V h_i. \tag{7}$$

The within-class scatter matrix S_w is defined as:

$$S_w = \sum_{j=1}^{k} \sum_{h_i \in C_j} (h_i - \mu_j)(h_i - \mu_j)^T. \tag{8}$$

The scatter matrix S_w captures how well-separated the clusters are in the embedding space. The transformation aims to project the embeddings into a new space where the cluster structure is more pronounced. The loss function for this transformation becomes:

$$L_3 = \mathrm{Tr}(V S_w V^T). \tag{9}$$

In this equation, V contains the eigenvectors of S_w, ordered by their corresponding eigenvalues. This orthonormal transformation reveals the important dimensions for clustering and reduces the influence of noise or irrelevant dimensions.

3.3.4. greedy optimization for enhanced clustering

To improve the clustering quality, we perform a greedy optimization step that minimizes the entropy of the data distribution in the transformed space. Specifically, we move data points closer to their centroids, particularly in the least informative direction, which corresponds to the eigenvector with the smallest eigenvalue:

$$L_4 = \sum_{j=1}^{k} \sum_{y_i \in C_j} \|y_i - y_i'\|^2, \tag{10}$$

Table 1
Public datasets used to evaluate our method.

Dataset	REUTERS-10K	20NEWS	RCV1-10K
Dimensions	(2000)	(2000)	(2000)
# of samples	210	2000	10 000
# of classes	4	20	4

Here, y_i' is the adjusted version of y_i, where its last dimension is modified to bring the point closer to its cluster centroid. This greedy optimization reduces the dispersion of points within each cluster, leading to tighter and more well-defined clusters.

3.3.5. joint optimization

The final step is the joint optimization of the entire model, which involves alternating between optimizing the representation learning (autoencoders) and the clustering objective (K-means). The combined loss function that we aim to minimize is:

$$L = L_1 + L_2 + L_3 + L_4. \tag{11}$$

By minimizing this total loss, we ensure that the learned representations are not only good at reconstructing the input data but also facilitate effective clustering. This alternating optimization iteratively updates the embeddings and clustering assignments until convergence, leading to better clustering performance.

4. Performance evaluation

4.1. Experimental setup

To demonstrate the superiority of our method among the state-of-the-art Deep Embedded K-Means Clustering method, we tested it to the same data used in Guo et al. (2021). These datasets are described in Table 1.

To assess the effectiveness of our method we employ external evaluation metrics, including Accuracy and Normalized Mutual Information (NMI). The Accuracy and NMI are external evaluation metrics that rely on ground truth labels to assess the agreement between the clustering results and the true cluster assignments, offering insights into the accuracy and alignment of the clustering method with the known data categories. Higher values of these indicators signify better clustering results, indicating the effectiveness of the method. Through this rigorous evaluation, we demonstrate the effectiveness and superiority of our method in identifying clusters for multi-density data.

Different representative methods are compared with our method, and are mentioned below:

- K-means clustering (Shukla and Naganna, 2014): Applying the k-means clustering algorithm to the raw datasets.
- PCA+K-means: Applying the K-means algorithm within the subspace defined by the initial p principal components of the data, obtained through Principal Component Analysis (PCA). The value of p is chosen to ensure that 90% of the data's variance is retained.
- AE+K-means: Executing K-means within the embedded space of the pre-trained convolutional/MLP autoencoder.
- DEC (Xie et al., 2016): It employs an MLP autoencoder to discover the embedding space and subsequently conducts clustering within this embedding space by minimizing the Kullback–Leibler (KL) divergence between the cluster distribution and a desired Student's t-distribution target.
- DCEC (Guo et al., 2017b): It substitutes the MLP autoencoder in the DEC framework with a convolutional autoencoder.
- IDEC (Guo et al., 2017a): It substitutes the MLP autoencoder in the DEC framework with an under-complete autoencoder designed to maintain the local structure of the data.

- DCN (Yang et al., 2017): It merges the objective function of the MLP autoencoder with that of K-means and alternately optimizes them.
- DKM (Fard et al., 2020): It views the K-means objective as a limit of a differentiable function and employs stochastic gradient descent to concurrently optimize representation learning and clustering.
- Deep Embedded K-Means Clustering method (DEKM) and ($DEKM_F$) (Guo et al., 2021)

For a fair comparison, the best performances for all competing methods are presented by directly reproducing the best experimental results from the corresponding published papers. We ensured that the parameter settings for each model followed those used in previous studies for fair comparison.

4.2. Experimental results

We conducted a comprehensive evaluation of our proposed model and compared it with other competitive methods using the datasets described earlier. The results osf these experiments are summarized in Table 2. The table showcases the best performance metrics, indicated in bold, which correspond to the highest values achieved.

To further emphasize the effectiveness of our solution, we conducted a comparative analysis between the results obtained using our method and those achieved by considering the individual views independently, referred to as "Proposed method$_{View1}$", which will be the same as $DEKM_F$, and "Proposed method$_{View2}$". This comparison, allowed us to assess the added value of utilizing complementary information from multiple views to enhance performance. The clustering performance of the single-view models, "Proposed method$_{View1}$" and "Proposed method$_{View2}$", is presented alongside the results of our proposed deep multi-view model in Table 2.

Based on the results presented in Table 2, it is evident that our proposed method surpasses other state-of-the-art techniques. These findings highlight the high efficiency and effectiveness of our solution, showcasing its superiority compared to existing methods.

Table 2
Clustering performance of different models (%).

Dataset	Method	ACC	NMI
REUTERS-10K	K-means	54.05	41.91
	PCA+K-means	53.97	41.35
	AE+K-means	70.52	49.02
	DEC	72.17	54.87
	DCEC	72.17	54.87
	IDEC	75.64	49.81
	DCN	–	–
	DKM	–	–
	DEKM	76.28	59.06
	DEKM_F	76.42	59.99
	Proposed method_View1	76.42	59.99
	Proposed method_View2	65.01	51.29
	Proposed method	**78.13**	**61.41**
20NEWS	K-means	21.59	19.70
	PCA+K-means	21.75	20.46
	AE+K-means	40.24	37.60
	DEC	35.65	43.79
	DCEC	35.65	43.79
	IDEC	40.50	38.20
	DCN	44.00	48.00
	DKM	51.20	46.70
	DEKM	41.08	40.27
	DEKM_F	41.20	43.76
	Proposed method_View1	41.20	43.76
	Proposed method_View2	40.86	40.32
	Proposed method	**41.63**	**45.43**
RCV1-10K	K-means	51.96	31.35
	PCA+K-means	51.91	31.39
	AE+K-means	64.11	41.12
	DEC	66.81	45.24
	DCEC	66.81	45.24
	IDEC	59.50	34.70
	DCN	56.70	31.60
	DKM	58.30	33.10
	DEKM	67.15	46.18
	DEKM_F	62.12	35.98
	Proposed method_View1	62.12	35.98
	Proposed method_View2	62.35	36.59
	Proposed method	**71.93**	**48.85**

5. Real-world case study — Air pollution

We further evaluate our method's performance using real-world application for one of the most critical and silent threats, the air pollution. This evaluation involves comparing our method not only with the DEKM method but also with the K-means clustering method, thereby providing a comprehensive benchmarking analysis.

5.1. Data collection

The first step of our process involves collecting air pollution data from remote sensing satellite. Additionally, we introduced supplementary data, including land use data, traffic flow, neighbors statistical data, to obtain different features of the data. Incorporating additional datasets helps in achieving a more comprehensive and accurate detection of different patterns of air pollution. This section provides a comprehensive overview of the dataset utilized in our study, encompassing various components crucial for our research. The dataset comprises Ground-based NO_2 measurements, Copernicus Sentinel-5P NO_2 maps, Land-use Data, and Traffic flow Data, each contributing essential information to our analysis. Below, we offer a detailed description of each dataset component, highlighting its significance and relevance to our investigation.

5.1.1. Ground-based NO_2 measurements

The European Environment Agency (EEA) offers a valuable resource for accessing air pollution data collected from fixed sensors. Through their database, users can download the necessary data. The EEA provides two types of data: E1a and E2a. The E1a data consists of information reported to the EEA by member states every September, covering the preceding year. For example, the data delivered in September 2022 encompasses the year 2021. In contrast, the E2a data includes up-to-date hourly data from various member states. For our research, we specifically utilize the E1a data, as it undergoes a thorough validation process and is considered the official delivery of air pollution measurements. This ensures the reliability and accuracy of the ground-based NO_2 measurements used in our study. Fig. 5 shows an example map of the data collected for the ground-based sensors. EEA provides a tool in a web service (EEA Authors, 2024) to help collect the data. Search criteria includes country, city name, pollutant, period and output format. Once specified, the required data can be directly downloaded. Another solution provided through this service consists of an auto-generated python script that could be executed in order to obtain the links of the files to download. Having obtained the required data, we conditioned and rasterized the data as explained in Section 5.2 (Data preprocessing).

5.1.2. Copernicus Sentinel-5P NO_2 maps

The Copernicus Sentinel-5P mission, launched on October 13, 2017, is a collaborative effort involving the European Space Agency (ESA), the European Commission, the Netherlands Space Office, industry partners, data users, and scientists. This mission includes a satellite

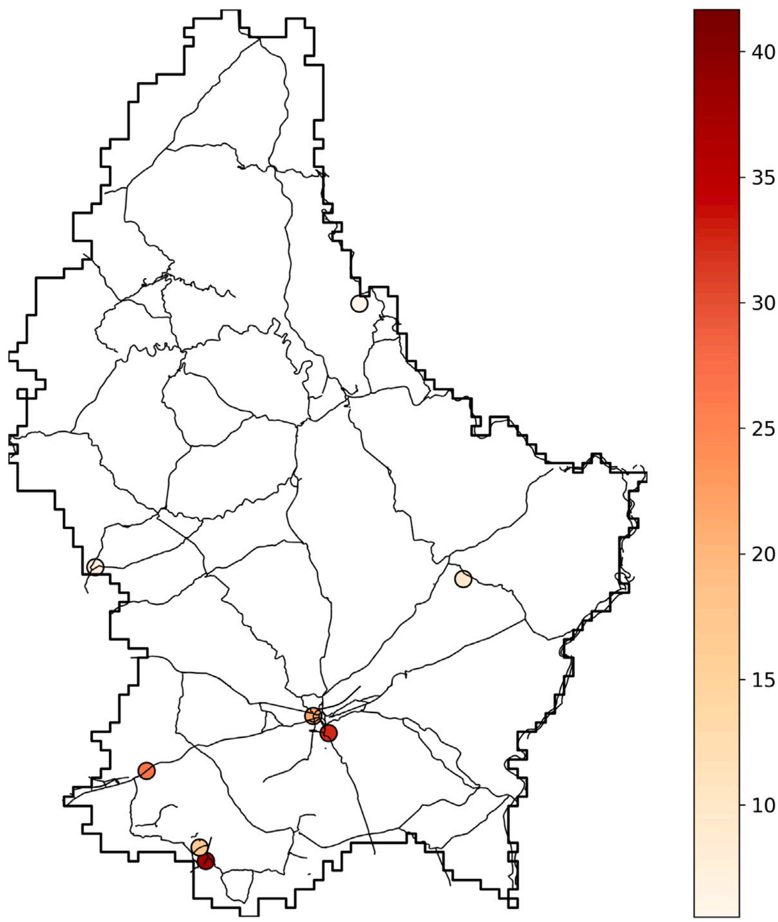

Fig. 5. Map of NO_2 ground-based sensors — Grand Duchy of Luxembourg.

equipped with the TROPOspheric Monitoring Instrument (TROPOMI), designed for high-resolution atmospheric measurements. Its primary objectives encompass air quality assessment, ozone monitoring, UV radiation analysis, and climate forecasting.

Access to Sentinel-5P user products is facilitated through the Copernicus Open Access Hub, which provides comprehensive, free, and open access to these valuable datasets. Additionally, the Copernicus Data and Information Access Services (DIAS) offer alternative platforms for data access (Copernicus Authors, 2023).

In our research, We utilized an open-source script that streamlines the download and preprocessing of air pollution data acquired from the Sentinel-5P mission. This script, detailed in Omrani (2023) and Omrani et al. (2020), demonstrated user-friendliness and exceptional convenience, significantly aiding our data collection process.

To obtain the average tropospheric NO_2 columns over Luxembourg for each month from 2018 until June 2021, we averaged the acquired maps. To convert the tropospheric column NO_2 abundance, measured in molecules per square centimeter (molecules/cm2), into ground-level NO_2 concentration expressed in parts per billion (ppb), we applied the Weather Research and Forecasting model coupled with Chemistry (WRF-Chem) using a standard methodology. This conversion allowed for meaningful comparison with the ground-based NO_2 measurements.

After collecting the air pollution data, the maps are cleaned and cropped for the study area as explained in Section 5.2 (Data preprocessing). Fig. 6 shows an example map of the resulting air pollution Sentinel-5P map.

5.1.3. Land-use data

For our study, we accessed land-use data from OpenStreetMap (OSM), a comprehensive and freely accessible database. OSM serves as a valuable resource, providing map data for numerous websites, mobile apps, and hardware devices. It is continuously updated and maintained by a global community of mappers who contribute information on various features, including roads, trails, cafés, railway stations, and more. The datasets we reprocessed from OSM include maps of primary roads, industrial zones, residential zones, and forest zones. These datasets offer rich and detailed information on land-use patterns, which greatly enhance the depth and accuracy of our analysis and research. Utilizing OSM data allows us to gain valuable insights into the spatial distribution of different land-use categories and their potential influence on capturing underlying air pollution patterns.

After collecting the various land-use features, the data is conditioned, cropped to the study area, and rasterized as explained in Section 5.2 (Data preprocessing). Fig. 7 shows an example map of the resulting map showing the various data collected for the country of Grand Duchy of Luxembourg.

5.1.4. Traffic flow data

The government of the Grand Duchy of Luxembourg provides access to traffic flow data obtained from around 178 traffic count sensors (until now) distributed across the entire country. These valuable traffic counts can be freely accessed through their websites and are made available in the form of Excel files. The sampling period for the data typically ranges from 2 to 3 hours.

To facilitate data analysis, the traffic flow measurements were downloaded and averaged. Subsequently, the data were converted into raster format to seamlessly integrate them into our model. Incorporating traffic flow data into our analysis adds a crucial spatial dimension, enabling us to explore the potential influence of traffic on capturing underlying air pollution patterns.

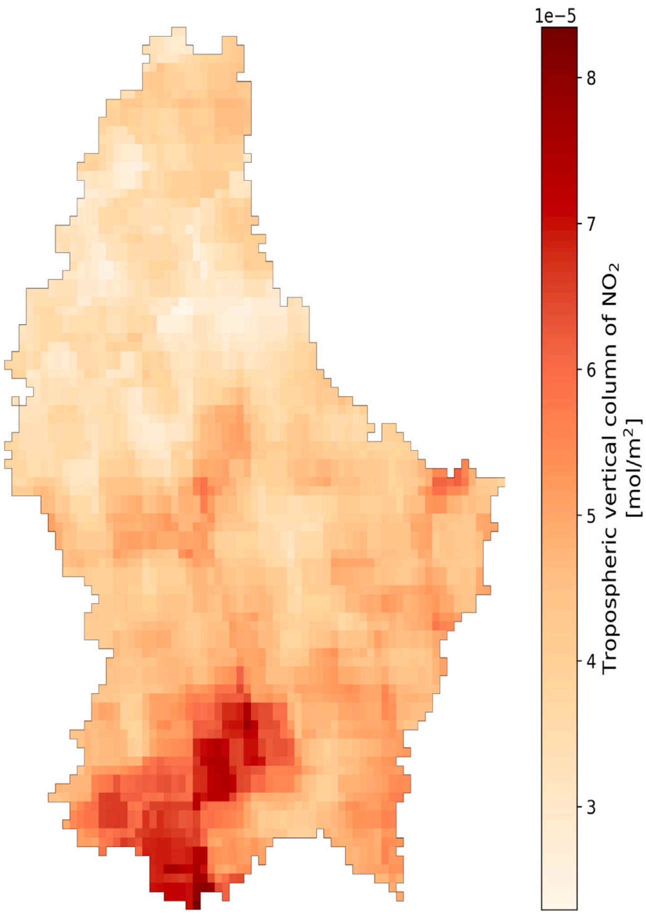

Fig. 6. Sentinel-5P NO_2 Map — Grand Duchy of Luxembourg.

After collecting the data, the traffic flow is first averaged by month. We then categorize the traffic points into three categories "high-traffic", "medium-traffic", "low-traffic" based on high and low thresholds set to 600000 and 120000 respectively. Having obtained the categorized traffic data, we calculate the distance to nearest high, medium, and low traffic points from the center of each rasterized cell within the study area. More data about rasterization can be found in Section 5.2 (Data preprocessing). Fig. 8 shows an example map of the resulting map showing the various data collected for the country of Grand Duchy of Luxembourg.

5.1.5. Neighbors effect

Recognizing that the air pollution level in a specific zone can be influenced not only by its internal characteristics but also by neighboring zones, we have taken into account the neighborhood effect. To capture this influence, we include a set of features that represent the average degree of air pollution in the neighboring zones, along with the maximum value and variance of pollution levels in those neighboring areas.

Incorporating the neighborhood effect is essential for our air pollution detection analysis, as it allows us to account for the spatial dependencies and interactions between adjacent zones. Further details and specifics of this approach are elaborated in the data preprocessing section, providing a comprehensive understanding of how we consider the neighborhood effect in our methodology.

5.2. Data preprocessing

After the data collection phase, the subsequent critical step is the data preprocessing phase, where the collected data undergoes various necessary transformations and preparations to be suitable for analysis. This phase involves cleaning the data, converting data types based on specific characteristics, rasterizing satellite images, performing convolution kernel operations, and preparing the data for subsequent steps. Furthermore, the preparation phase includes the extraction and processing of additional sub-features, such as statistical data, to enhance the comprehensiveness of the input data. Table 3 summarizes the final pre-processed features used in our proposed model, showcasing the diverse set of inputs we leverage to capture air pollution patterns.

Table 3
Description of the features employed.

Feature	Buffer size
Distance to the nearest primary road	–
Distance to the nearest high traffic	–
Distance to the nearest moderate traffic	–
Distance to the nearest low traffic	–
Industrial zone	–
Neighbor industrial zones (Buffered)	5 km
Neighbor industrial zone Variance	5 km
Neighbor industrial zone Distance to mean	5 km
Residential zone	–
Neighbor residential zones (Buffered)	5 km
Forest zone	–
Neighbor forest zones (Buffered)	5 km
Neighbor forest zone variance	5 km
Neighbor forest zone distance to mean	5 km
Neighbor NO_2 average value (Buffered)	5 km
Neighbor NO_2 Variance	5 km
Neighbor NO_2 Distance to mean	5 km

5.2.1. Cropping

The cropping phase in data preprocessing is critical for refining and optimizing datasets, particularly in contexts involving images or spatial data, which is the case in our application. By isolating the study area or region of interest (ROI), cropping allows the analysis to focus on relevant information while discarding irrelevant or distracting elements. This not only reduces the overall size of the dataset, making it more manageable and efficient to process, but also enhances the accuracy of models by eliminating noise and irrelevant data that could confuse the analysis. Cropping ensures consistency across datasets, especially when integrating multiple sources, and improves the clarity of data visualizations by removing unnecessary details.

5.2.2. Cleaning

Cleaning is a crucial step in data preprocessing, involving the identification and rectification of errors, inconsistencies, and missing values in a dataset. Data cleaning step in our case included handling missing data by deleting data records with no-data values. Standardizing data formats is another key task, ensuring that all data points are in a consistent format, such as date formats, units of measurement, joining categorical data (e.g. land-use) and numerical ones (e.g. traffic) . This step is crucial for integrating data from multiple sources. Without proper cleaning, all our subsequent steps could lead to misleading results.

5.2.3. Base grid

A base grid is crucial when working with multi-source and multi-format data that needs to be rasterized such as the case of air pollution problem (i.e. data includes satellite imagery, sensor networks, land-use, each using different coordinate systems, resolutions, and formats). It provides a standardized spatial framework onto which all data can be projected, processed, and merged. A base grid simplifies data integration by offering a common reference frame, enabling efficient overlay and merging of diverse datasets. It minimizes misalignment errors, while also optimizing computational processes.

The creation of the base grid involves several steps that ensure it serves as a standardized spatial reference for aligning, integrating, and

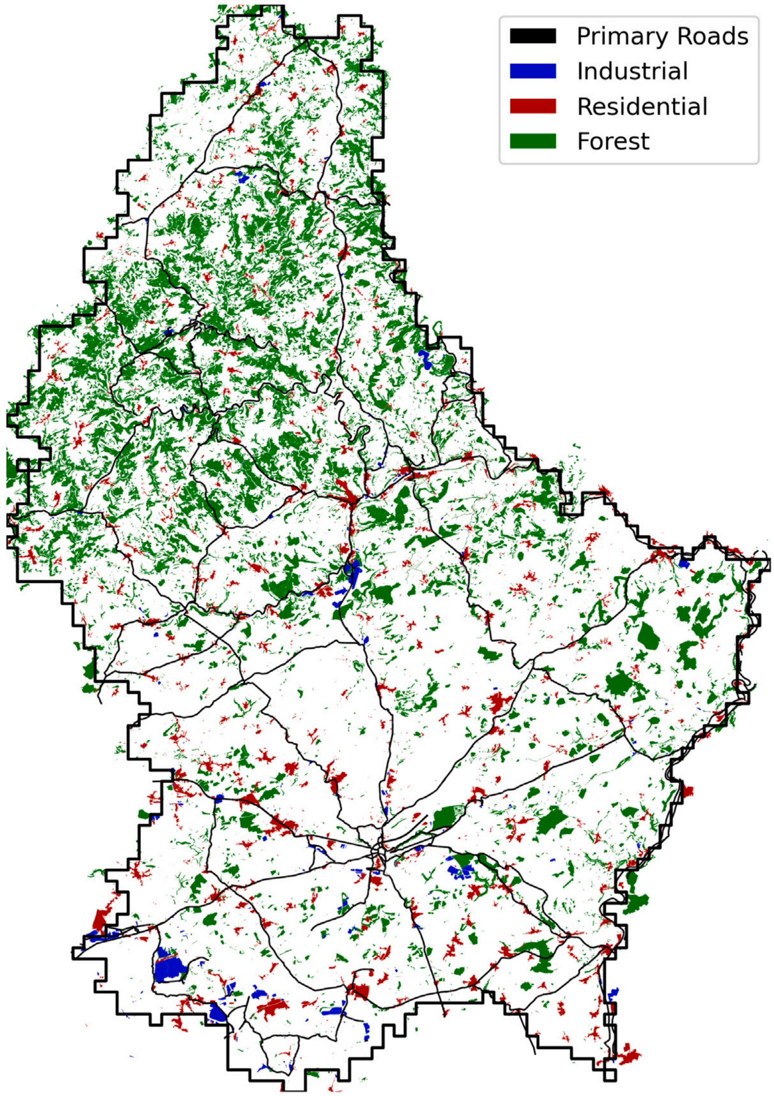

Fig. 7. Land-use Map — Grand Duchy of Luxembourg.

analyzing the data. First step is to define the extent of our study area, this means determining the geographic area that the base grid will cover. Second step is to select a coordinate system, we have chosen the World Geodetic System 1984 ensemble (WGS 84) for our application. Then comes defining the grid resolution, this was based on the lower resolution in our dataset which is the resolution of the Sentinel-5P air pollution maps (5.5 kmx3.5 km). The final step is to create the grid structure, this is done based on the defined extent, coordinate system, and resolution. The final base grid consists of evenly spaced cells, each representing a specific geographic area. Later preprocessing steps will make sure that the whole dataset will be projected to this base grid.

5.2.4. Rasterization

Rasterization involves transforming input data, such as vector maps or shape representations, into pixelated images or bitmaps, allowing for consistent and compatible data representation. This technique, originally utilized in early television technology, has been widely adopted in modern display systems to convert electronic data or signals into projected images, including video or still graphics (Worboys, 1995). Satellite images are typically provided in a rasterized format, while other features may be in free-form formats, necessitating the process of rasterization.

In real-world application, such as air pollution, rasterization is crucial because many spatial analysis tools and algorithms are optimized for raster data, enabling efficient execution of tasks such as land use classification. The grid format of raster data simplifies operations such as overlay analysis, proximity calculations, and spatial filtering. Additionally, rasterization facilitates compatibility between the multi-source dataset, ensuring that complex spatial analyses can be conducted seamlessly. With the help of the rasterization step, the multi-source data records are projected into the base grid previously defined.

5.2.5. Nearest neighbor

The nearest neighbor concept is a fundamental approach in data analysis and machine learning used to classify or predict outcomes based on the proximity of data points. This method involves identifying the closest data points to a given query point within a feature space. We have used the K–d tree concept, which is used to organizes multidimensional data points to facilitate efficient searches. It partitions the space using hyperplanes along different dimensions, creating a binary tree where each node represents a k-dimensional point. During a nearest neighbor search, the k–d tree narrows down the search space by traversing the tree and examining regions where the closest points are likely to be. This method significantly speeds up the search compared to a brute-force approach by pruning large areas of the data.

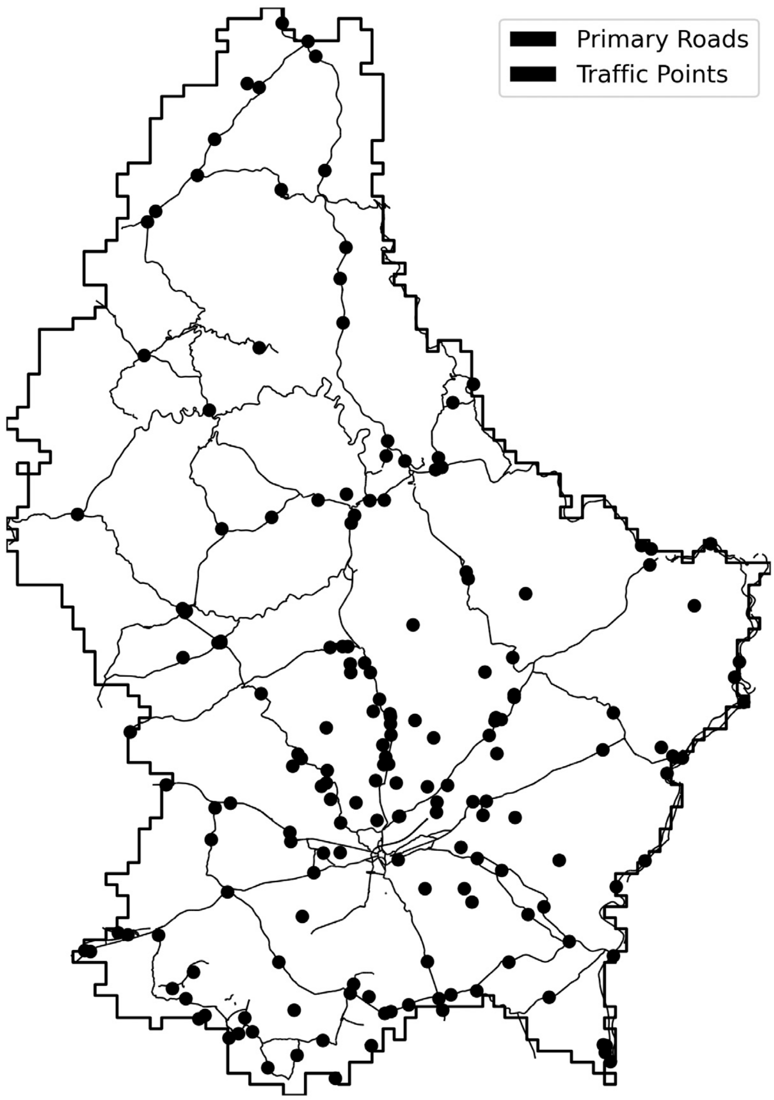

Fig. 8. Land-use Map — Grand Duchy of Luxembourg.

This is mainly used in our application to define the nearest traffic points and sensors to the base grid's cells.

5.2.6. Convolution kernel

Convolution Kernel Operation (CKO) is an advanced image-processing technique that leverages multi-pixel operations using convolution kernel masks. This is a fundamental component in data preprocessing, particularly in image processing and computer vision tasks. This process modifies each output pixel based on contributions from neighboring input pixels, enabling various operations. We have used the CKO to compute the average kernels corresponding to each cell for land-use data, NO_2, and neighbors effect.

5.3. Evaluation

We evaluate the performance of our method using the collected real-world air pollution data for Luxembourg. This dataset contains 3765 samples, each consisting of 13 features. A sample is shown in Table 4.

As we do not have ground truth labels for the air pollution data, we utilize the Silhouette score as the evaluation metric. The Silhouette score measures the compactness and separation of clusters based on the distance between data points and their cluster centroids, providing an indication of the clustering quality without the need for ground truth labels. By assessing how well each data point fits within its allocated cluster, the silhouette measures the compactness and spacing of clusters. The average Silhouette coefficient over all data points is used to determine the Silhouette score for a certain set of clusters. Better-defined and well differentiated clusters are indicated by higher silhouette ratings. We calculated the Silhouette scores for various number of clusters and plotted the outcomes on Table 5. As shown in Table 5, the proposed multi-view method outperform all other methods including the proposed single-view method. This is mainly due to the fact that multi-view methods ensures having complementary information. For instance, each view provide unique information, different views might highlight different aspects of the data, leading to better features representation. Models with multi-views are also more robust to noise and errors. If one view contains noise or errors, the other views can help compensate. Another benefit of having multi-view model is that it makes more efficient use of available data. Multi-view model can learn from multiple sources simultaneously, which can be especially useful when each view alone might be insufficient or sparse.

Another common challenge in clustering techniques is to determine the ideal number of clusters. For this matter, we employed the Elbow technique to further analyze the outcomes of the clustering steps in order, which helps in determining the ideal number of clusters for our deep K-means clustering method. For various values of K (the

Table 4
Sample from our dataset.

Distance to primary road	Distance to nearest high traffic point	Distance to nearest medium traffic point	Distance to nearest low traffic point	Landuse industrial variance	Landuse industrial distance to mean	Landuse industrial buffered (5 km)	Landuse residential buffered (5 km)	Landuse forest buffered (5 km)	Neighbors NO2 variance	Neighbors NO2 distance to mean	Neighbors NO2 distance to max	NO2 Value
0.0964	0.1276	0.1101	0.2885	0.6573	0.1500	0.0000	0.2500	0.0000	0.2270	0.0239	0.5950	0.3730
0.1140	0.0728	0.1562	0.2524	0.6023	0.0000	0.0000	0.5000	0.0000	0.2690	0.2150	0.6840	0.4290
0.0828	0.0355	0.1422	0.2163	0.6820	0.0500	0.0000	0.2500	0.0000	0.2890	0.2660	0.5280	0.4490
0.0803	0.0612	0.0995	0.1803	0.5914	0.9500	0.0000	0.0000	0.1429	0.2610	0.3300	0.6520	0.4720
0.0536	0.0901	0.0657	0.1444	0.6023	0.2500	0.1667	0.2500	0.1429	0.2760	0.3160	0.4800	0.4890
0.0014	0.0455	0.0586	0.1087	0.5624	0.9500	0.1667	0.2500	0.1429	0.2100	0.2550	0.4700	0.4950

Table 5
Clustering performance of the single view and multi-view models applied to air pollution data (%)..

Model	K-means	DEKM	Proposed single-view method	Proposed multi-view method
Silhouette	28.29	50.48	52.76	61.76

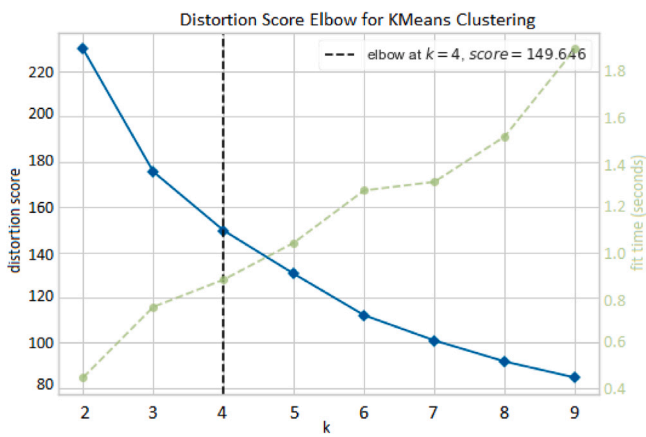

Fig. 9. Elbow method applied to our proposed method.

number of clusters), it entails visualizing the sum of squared distances (inertia) between data points and their respective cluster centroids. The plot's Elbow point, which symbolizes the point at which the inertia is no longer appreciably decreased by the addition of more clusters, corresponds to the ideal value of K. The relationship between the quantity of clusters and the accompanying inertia values, in addition to the time complexity (s), was examined using the Elbow approach. We determined the ideal number of clusters by looking at the Elbow point in the plot, which achieves a compromise between reducing inertia and preventing overfitting. The obtained plot is illustrated on Fig. 9. According to this figure, the ideal number of clusters, corresponding to our dataset, is equal to 4.

5.4. Clusters visualization

To gain deeper insights into the clustering results obtained by adopting our deep multi-view embedded k-means method, we performed two-dimensional (2-D) and three-dimensional (3-D) visualizations of the obtained clusters. These visualizations offer a clear representation of the distribution of data points within the clusters, which reflect the underlying structure captured by our proposed clustering algorithm.

In our visualizations, we applied two widely used dimensionality reduction methods: Principal Component Analysis (PCA) (Jolliffe, 2003) and t-Distributed Stochastic Neighbor Embedding (t-SNE) (Van der Maaten and Hinton, 2008). PCA is designed to preserve the maximum variance of the data within a reduced-dimensional space, whereas t-SNE focuses on maintaining pairwise distances between data points to reveal intricate relationships in the data.

The resulting 2-D and 3-D visualizations are shown in Figs. 10, 11, 13, 13. These figures effectively illustrate the separation and coherence of the clusters. PCA visualizations, presented in Figs. 10, 11 allow to explore the variance, structure, and separability in our high-dimensional data by projecting it into a space that is easier to interpret, 2D and 3D respectively. It is clear that the first axis ('pca_1' in Fig. 10 and 'x' in Fig. 11) accounts for the majority of the variance, while a non-linear structure is evident in these lower dimensions, as data points do not align along straight planes. The clusters are distinctly separated with few outliers, suggesting that most of the data fall into well-defined clusters with clear boundaries. Fig. 11 provides a more complete picture of the underlying structure. Although the clusters are well-separated in Fig. 10, the third dimension ('z') contributes additional meaningful information, explaining a relatively similar amount of variance as the second axis ('y'). Despite this, the PCA 3-D visualization confirms the high separability of the clusters observed in the PCA 2-D visualization.

Given that the PCA 2-D and 3-D visualizations reveal a non-linear structure and PCA is limited to capturing linear relationships by focusing on directions of maximum variance, it was worth applying the TSNE 2-D and 3-D visualizations (presented in Figs. 12 and 13) to gain a deeper understanding of the local structure, shapes, and densities of each cluster. It is worth noting that clusters 1 and 3 exhibit tighter grouping (indicated by the high-density regions) compared to 0 and 2 which shows more sparse regions. This suggests that points within clusters 1 and 3 are more similar to each other. TSNE visualization results align with PCA in terms of clusters separability, clearly distinguishing points from different clusters. However, there are few exceptions within clusters 0 and 2, where certain points appear to share similar characteristics. Fig. 13 provides additional insight into the volume of clusters in 3-D, illustrating the spatial extent of each group in the lower-dimensional space. In this visualization, cluster 0 is occupying the largest space.

This visual assessment complements the quantitative metrics and enhances our understanding of how effectively our proposed method captures patterns and characteristics within the air pollution data.

5.5. interpretation

Upon establishing that our proposed method demonstrates strong performance in terms of clustering separability, we shifted our focus to a detailed analysis of the resulting distributions to extract meaningful insights. Recognizing that air pollution is influenced by several factors such as traffic and land use types (e.g., industrial, forested zones), we conducted a practical analysis to evaluate whether the clustering results align with theoretical expectations.

As depicted in Fig. 14, Cluster 0 primarily includes points that are located at a significant distance from high and medium traffic areas. Conversely, Cluster 1 encompasses points that are situated very close to, or in proximity to, primary roads and high traffic zones, as well as those near medium traffic areas. The majority of points in Cluster 3 are positioned near primary roads and high or medium traffic zones. Cluster 2, however, exhibits a distribution spread across all defined categories. In this context, "very close" is defined as a distance of less than 200 meters from traffic points, while "close" or "in proximity

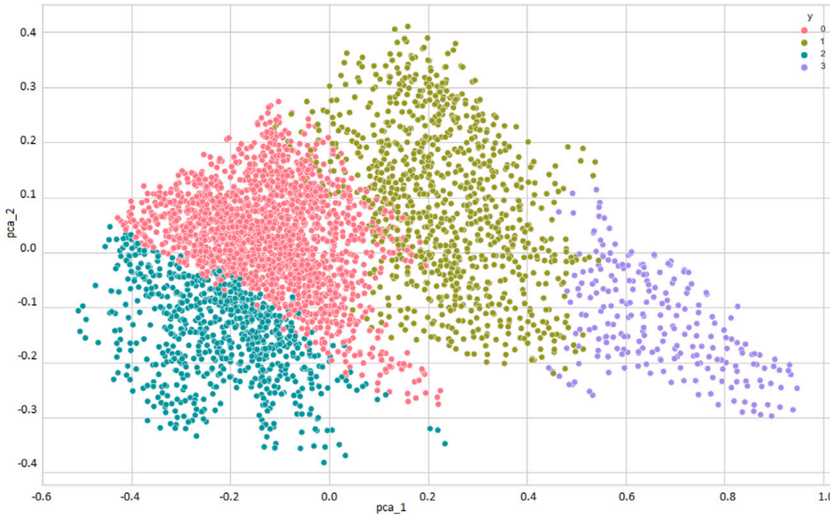

Fig. 10. PCA 2D visualization of the predicted air pollution clusters.

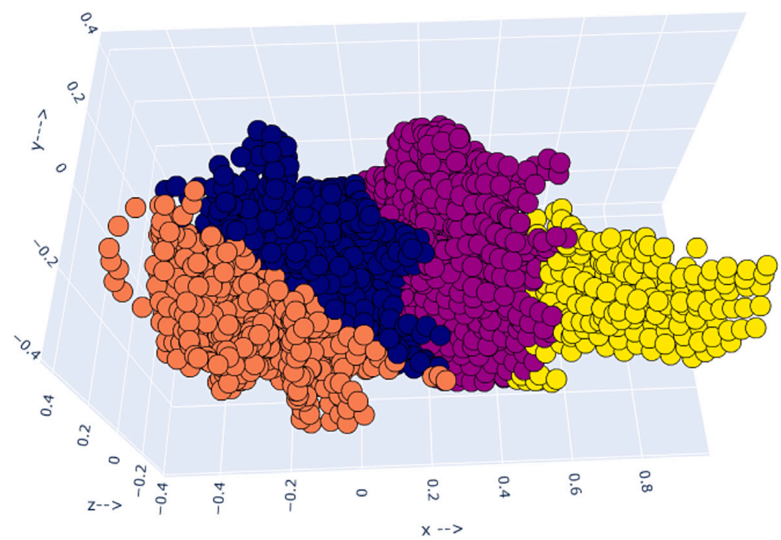

Fig. 11. PCA 3D visualization of the predicted air pollution clusters.

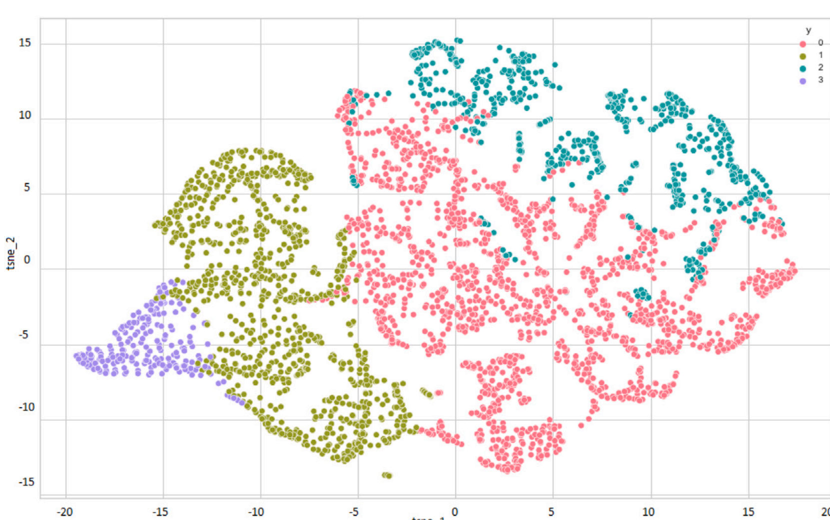

Fig. 12. TSNE 2D visualization of the predicted air pollution clusters.

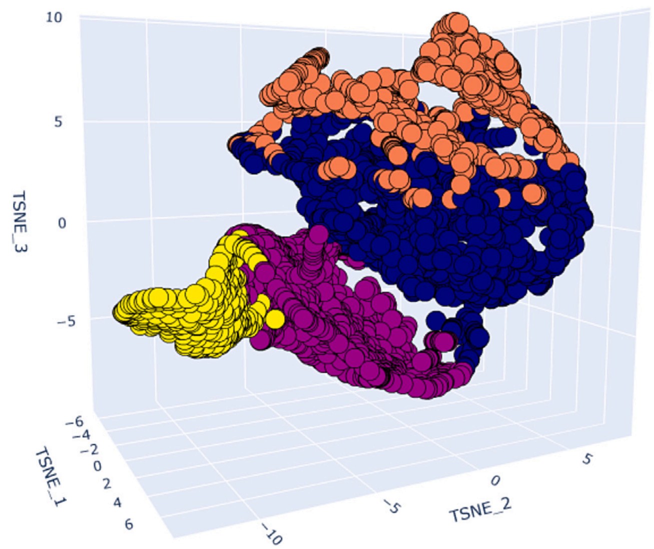

Fig. 13. TSNE 3D visualization of the predicted air pollution clusters.

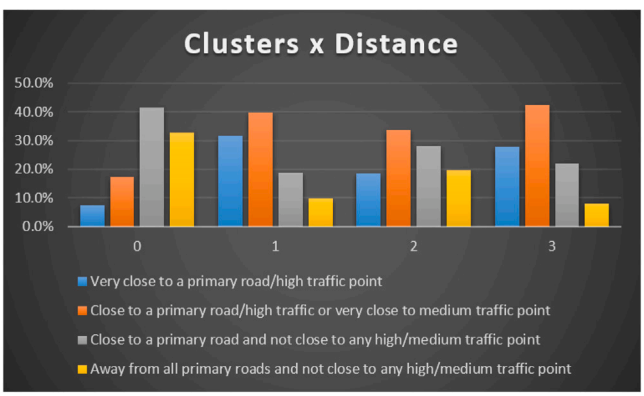

Fig. 14. Clusters distributions with respect the closeness to primary roads and traffic points.

to" refers to distances between 200 and 1000 meters. Traffic points classification is explained in Section 5.1.4 (Traffic Flow Data).

Fig. 15 illustrates the distribution of land-use types (described in Section 5.1.3 — Land-use Data) within the clusters. It is evident that forested areas are predominant within Cluster 0, whereas residential areas are the main contributors to Cluster 1. Industrial areas are most prevalent in Cluster 3, while Cluster 2 again shows no distinct distribution pattern. Areas were classified as forest, industrial, or residential based on their local land-use type and the land-use type within a 5 km buffer zone. If an area lacks a specific land-use type, the dominant feature within the buffer determines its classification.

Figs. 14 and 15 complement each other, indicating that Cluster 0 likely represents areas with lower air pollution due to the abundance of forest zones. In contrast, Cluster 3 predominantly includes industrial zones, which may correspond to areas with higher air pollution levels. Cluster 1 appears to consist primarily of residential zones, which could also contain areas with medium to high air pollution levels.

Fig. 16 presents the percentage of points within each cluster, highlighting that Clusters 0 and 1 account for the majority of the points.

To validate the conclusions drawn from these analyses, we visualized the data using Box and Whisker plots for all clusters, as shown in Figs. 17, 18, 19, 20. The results suggest that Clusters 1 and 3 encompass the majority of areas with high levels of air pollution, while Cluster 0 is predominantly composed of areas with low air pollution levels.

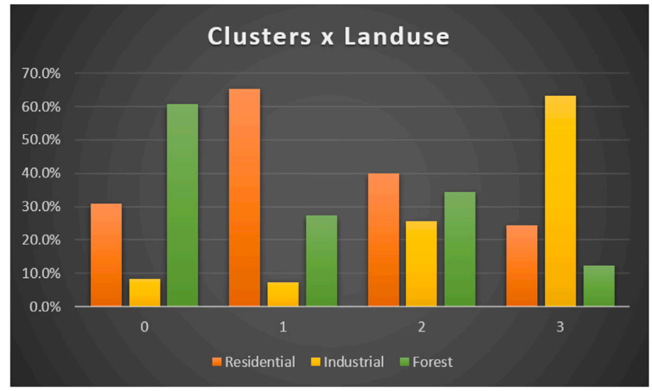

Fig. 15. Clusters distributions with respect land-use area types.

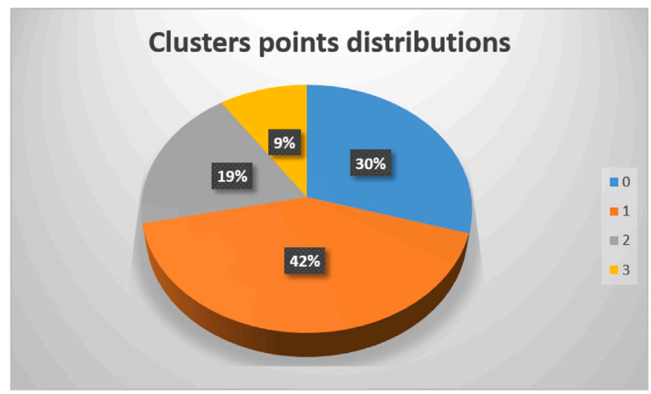

Fig. 16. Clusters points distributions.

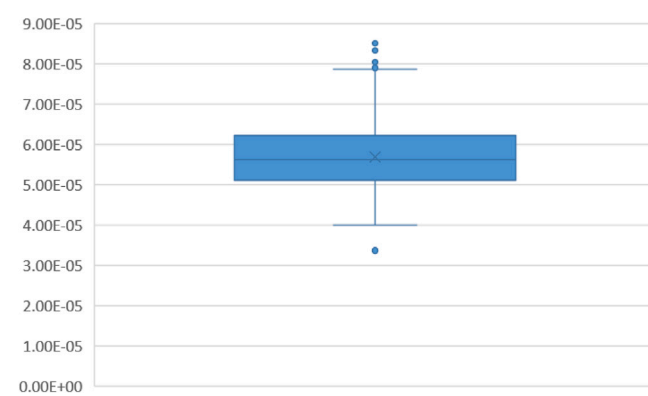

Fig. 17. Box and Whisker Plot for the NO_2 values distribution of points belonging to cluster 0.

Cluster 2, on the other hand, exhibits average pollution levels, making it challenging to derive significant insights from this cluster.

As concluded from previous analysis, Clusters 1 and 3 contain the majority of areas with high levels of air pollution, whereas Cluster 0 primarily consists of areas with low air pollution. Cluster 2, however, displays average pollution levels, making it difficult to extract meaningful insights from this cluster.

6. Conclusion

In this paper, we have introduced a novel multi-view deep embedded clustering method for multi-density applications. Our method aim

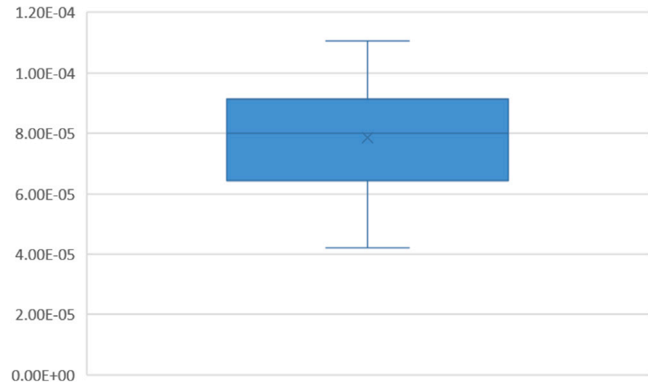

Fig. 18. Box and Whisker Plot for the NO_2 values distribution of points belonging to cluster 1.

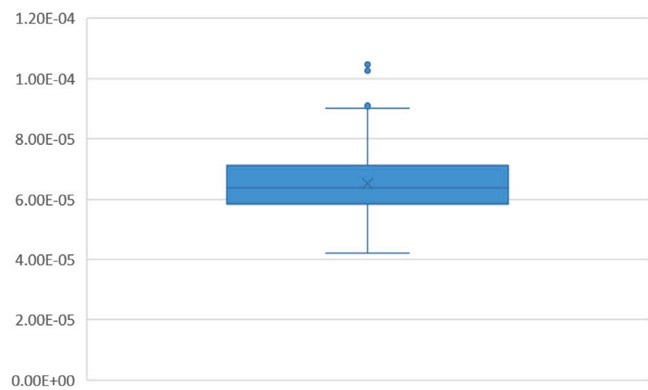

Fig. 19. Box and Whisker Plot for the NO_2 values distribution of points belonging to cluster 2.

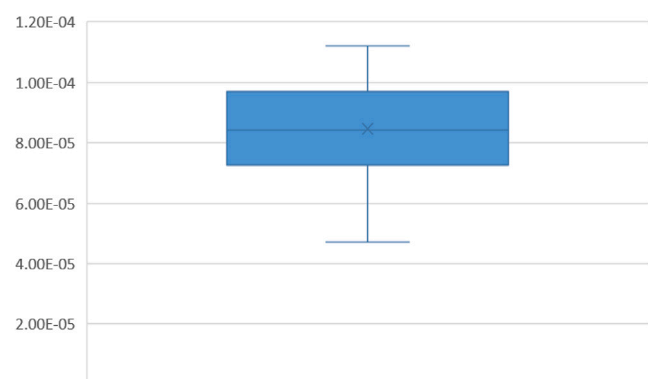

Fig. 20. Box and Whisker Plot for the NO_2 values distribution of points belonging to cluster 3.

to solve the problem of conventional density-based clustering methods that have shown limitations in handling multi-density areas. Our proposed method leverages the power of deep learning and employs an autoencoder to generate an embedding space. To reveal cluster-structure information, we further transform this space using an orthonormal matrix comprising the eigenvectors of the within-class scatter matrix of K-means. This transformation contributes significantly to the accuracy and efficiency of detection hidden air pollution patterns. Moreover, the adaptability of our method to real-world application such as the air pollution demonstrates its applicability in diverse environmental settings, thereby making a significant stride towards sustainable and healthier living conditions for communities affected by air pollution by aiding in informed decision-making processes for pollution management and urban planning. By conducting comprehensive experiments in the context of Luxembourg, we have demonstrated superiority in performance compared to existing state-of-the-art methods. Moreover, its efficiency and versatility are also validated through comprehensive testing on diverse datasets, including various publicly available datasets. This method could thus be adapted to many other real-world problems with little efforts. It remains to be determined how well this outcome can be integrated within machine learning models that predict air pollution level and detect hotspots. Furthermore, it would be interesting to study different models within the proposed architecture in a way to enhance the results of this method.

CRediT authorship contribution statement

Hassan Kassem: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Data curation, Conceptualization. **Sally El Hajjar:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Fahed Abdallah:** Writing – review & editing, Validation, Conceptualization. **Hichem Omrani:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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